Manikandan Valliappan

CFD Simulation Engineer

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Date of Birth	18.01.1996
Nationality	Indian
Visa Type	Permanent Residence
Languages	English – C1, German – B1, Tamil – Native
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PROFILE

CFD Simulation Engineer specializing in OpenFOAM-based solver development, multiphase flow modelling, and numerical validation for engineering flow problems. Developed and benchmarked CFD solver extensions across OpenFOAM 10–13, including C++/MPI-based coupling, timestep synchronization, volume-fraction treatment, and parallel data exchange. Achieved less than 1% terminal-velocity error in validated single-particle settling simulations, with strong experience in CFD validation, solver debugging, Python/C++ automation, Linux, MPI, and HPC-based workflows.

TECHNICAL SKILLS

CFD Simulation: OpenFOAM, ANSYS Fluent, COMSOL Multiphysics, STAR-CCM+, Multiphase Flow

Solver Development: C++ solver modification, numerical methods, validation, benchmarking, debugging

Programming: Python, C++, MATLAB, LaTeX

FEM Simulation: ANSYS Workbench, ANSYS Mechanical APDL, ABAQUS

Design Software: SOLIDWORKS, AutoCAD, CATIA V5

WORK EXPERIENCE

Research Associate — **09/2022 – Present**

Technische Universität Clausthal — *Clausthal-Zellerfeld, Germany*

- Developed and benchmarked OpenFOAM-based CFD solver extensions for incompressible, multiphase, porous-media, particle–fluid interaction, and rock-boring simulation applications related to tunnel-boring-machine interaction with granular media.
- Implemented C++/MPI-based coupling features for parallel data exchange, timestep synchronization, force coupling, volume-fraction coupling, and distributed particle search in coupled CFD simulations.
- Validated solver accuracy against analytical and benchmark cases, achieving less than 1% terminal-velocity error in single-particle settling simulations.
- Conducted cross-version benchmarking and parallel-consistency checks across OpenFOAM 10–13 to evaluate solver robustness, numerical accuracy, and scalability.
- Improved coupled-solver stability by resolving timestep-control, force-exchange, volume-fraction, boundary-condition, and MPI communication issues during parallel CFD runs.
- Automated repetitive simulation post-processing tasks using Python and C++, replacing manual data extraction from solver output files with reproducible workflows for error analysis, trajectory tracking, velocity comparison, and visualization.
- Built validation workflows for single-particle settling and packed-bed pressure-drop cases to compare CFD results against analytical and reference solutions.

Research Assistant — **06/2021 – 08/2022**

Fraunhofer Institute for Photonic Microsystems IPMS — *Dresden, Germany*

- Developed ANSYS APDL-based FEM simulation scripts for prismatic and S-shaped clamped-clamped Nano Electrostatic Drive beam structures.
- Performed nonlinear static, modal, and linear perturbation analyses to evaluate pull-in voltage, charge, displaced area, and electromechanical resonance frequency.
- Conducted parametric FEM studies covering up to 1,875 prismatic-beam and 625 S-shaped-beam geometry combinations.
- Applied Buckingham Pi theorem and regression analysis to derive empirical relationships between geometrical parameters and electromechanical performance variables.
- Automated CSV-based post-processing using Python to extract selected parameter sets, fit regression models, and compare FEM results with analytical solutions.

Chassis Design Engineer – Formula Student — **11/2018 – 06/2019**

Racetech Racing Team — *Freiberg, Germany*

- Designed chassis and rollover-hoop components for the RT13 race car using SOLIDWORKS sheet-metal modelling.
- Collaborated with a multidisciplinary team on structural layout, manufacturability, packaging, and vehicle-design constraints.

EDUCATION

Doctoral Candidate in Geomechanics and Multiphysics Systems — **09/2023 – Present**

Technische Universität Clausthal — *Clausthal-Zellerfeld, Germany*

- Doctoral research: *Rock Boring Simulation: A Coupled Approach with OpenFOAM and YADE.*
- Focus areas: CFD–DEM coupling, OpenFOAM–YADE solver development, particle–fluid interaction, and numerical validation.

M.Sc. Computational Materials Science — **10/2018 – 03/2022**

Technische Universität Bergakademie Freiberg — *Freiberg, Germany*

- Master's thesis: *Parametric Study of the Nano E-Drive Mechanism in Clamped-Clamped Beams*, carried out at Fraunhofer IPMS.
- Focus areas: Computational fluid dynamics, FEM analysis, numerical simulation, and material modelling.

B.E. Mechanical Engineering — **07/2013 – 04/2017**

Anna University — *Chennai, India*

- Final-year project: *Process-time reduction by introducing an automatic nut-runner in a commercial vehicle manufacturing environment*, carried out at Ashok Leyland Ltd.
- Focus areas: Fluid mechanics, thermodynamics, mechanical design, CAD/CAE, and engineering analysis.

SELECTED RESEARCH OUTPUT

- Manuscript in final revision for *Open Geomechanics*: *Numerical Benchmarking of CFD–DEM Coupling in OpenFOAM–YADE Using Stokes Settling.*
- Manuscript in final revision for *Open Geomechanics*: *Hindered Settling Simulation Using OpenFOAM–YADE CFD–DEM Coupling.*
- Manuscript in final revision for *Open Geomechanics*: *Ergun-Equation-Based Pressure-Drop Validation for Packed-Bed Flow Using OpenFOAM–YADE CFD–DEM Coupling.*